Causal ECG replay with a persistent patient twin

Runtime sees only the past; labels and scoring live offline. Manufactured-data UI demo.

ONLINE PATH (causal): no future samples, annotations, onset or pair identifiers

Replay service

- Historical ECG on a 30 s tick
- Single worker, explicit ticks

→

Causal features

- Trailing 10 min of ECG
- RR variability, signal quality

→

Frozen predictor

- Logistic: age/sex + ECG features
- Dev-only calibration; slope guard

→

Alert state machine

- Persistence 60 s, hysteresis
- Cooldown 600 s; abstains on poor quality

→

SQLite twin store

- One atomic commit per step
- Cursor, baseline, warning

→

FastAPI + React UI

- ECG, trends, risk, quality
- Warning history, clinician review

per-patient baseline updated AFTER each prediction

OFFLINE PATH (labels and scoring): never imported by runtime code

Data audit + labels

- IRIDIA-AF and LTAF WFDB; future rhythm intervals
- Cohort feasibility gate

→

Training + freeze

- Patient-level splits; fit / stop / calibrate / policy groups disjoint
- Hashed artifact + manifest

→

Complete-stream evaluation

- Whole recordings in time order; misses kept
- One-to-one matching; FAR per eligible hour

Shared by runtime and evaluation

- Alert state machine and evaluation code are one code path, so scored behaviour equals served behaviour.
- Default demo: manufactured data and a synthetic-trained model. Opt-in *serve --research-model* runs the gate-failed frozen finalist behind a red FAILED banner.

Model families tried (all offline, all development-only)

- Logistic and boosted classical models; RR-TCN16 (RR-interval CNN); waveform CNN with causal TCN head.
- Frozen ECG foundation-model (ECGFounder) attempt: no outcome results.

Source: README.md Architecture; reports/ml-investigation/report.md; frozen-challenge-fusion/config.json (persistence 60 s, cooldown 600 s).

Evidence and boundaries, at a glance

What was frozen, what was measured, and what is not claimed.

Release and evaluation boundaries

- **Development only:** 121 IRIDIA-AF patient groups; fit, stop, calibration and policy groups disjoint.
- **Freeze before opening test:** model, calibration, schema, policy, split and seed hashed before primary and LTAF were scored. No held-out retuning; both cohorts are now used up.
- **Complete chronological evaluation:** misses, false warnings, occupied time, coverage reported together. LTAF record groups do not prove independent people.
- **Local deployment:** localhost FastAPI + React + SQLite; manufactured-data demo; no hospital authentication or PHI governance.

Measured results (event level)

- Frozen age/sex + ECG logistic finalist detects **6/241** development, **1/43** primary held-out, **2/29** LTAF events (sensitivity 2-7%).
- Against random timing the development result is within chance (6 observed vs about 4.5 expected; P(at least 6) about 0.33).
- LTAF false-warning rate 0.008/h is not skill: 392 of 398 warning starts fall in unassessable time.
- The age/sex inputs show no benefit: ECG-only reference 7/241 vs fused 6/241.

Defect found and guarded

- Fold-0 calibrators learned negative slopes and reversed ranking (AUROC 0.573 raw to 0.427).
- A slope guard now raises in train.py::_calibrate; test in tests/test_calibration_slope.py. Frozen models had positive slopes, so held-out results are unaffected.

Not claimed

- No clinical readiness, no fusion benefit, no Indian-population evidence.
- Research criteria (0.25 false warnings/h, 50% sensitivity) are ours, not the organizers'; the acceptance gate fails.

Source: reports/ml-investigation/{report.md, frozen-cohorts, reference-evaluations}; reports/ml-investigation/chance-analysis/*.json.